

Web Technologies

Unit 1: Foundations of Internet and World Wide Web

1. Introduction to the Internet

Definition

- The Internet is a worldwide collection of interconnected computer networks that communicate using TCP/IP protocols to share information and resources.
- It enables computers and other devices worldwide to exchange information and access services such as websites, email, file transfer, online banking, cloud computing, and video conferencing.
- The Internet is often called the “Network of Networks” because it connects millions of private, public, academic, business, and government networks into a single worldwide communication system.

Services Provided by the Internet

- World Wide Web (WWW)
- Electronic Mail (E-mail)
- File Transfer Protocol (FTP)
- Video Conferencing
- Cloud Computing
- Online Banking
- E-commerce
- Social Networking
- Online Education

Network

A network is a collection of two or more computers and other devices connected together to communicate and share data, resources, and services.

Definition: A computer network is a group of interconnected computers and devices that exchange information using communication links and networking protocols.

2. Transaction Flow in the Internet

Example: Suppose a student wants to open the website www.college.edu to check examination results. The complete flow is as follows:

Step 1: Computer

The user opens a web browser and enters the URL www.college.edu. On pressing Enter, the browser creates an HTTP/HTTPS request for the homepage.

Step 2: Modem

The request is sent to the modem, which converts the computer's digital data into signals that can travel through the Internet connection (and converts incoming signals back into digital data).

Step 3: ISP (Internet Service Provider)

The modem forwards the request to the ISP, which provides Internet connectivity to users (e.g., Jio Fiber, Airtel Xstream Fiber, BSNL Fiber, ACT Fibernet).

Step 4: DNS (Domain Name System)

The ISP contacts the DNS server, which converts the website name into its IP address (e.g., www.college.edu → 192.168.20.15), so the browser knows where the website is located.

Step 5: Internet

Using the IP address, the request travels through the Internet — millions of interconnected networks of routers, switches, and cables — until it reaches the destination data center.

Step 6: Data Center

The request reaches the data center where the website is hosted — a large facility containing thousands of servers operating continuously to provide online services.

Step 7: Router

Inside the data center, a router receives the request, examines the destination, and forwards it to the correct server.

Step 8: Web Server

The web server receives the request, processes it, and determines which webpage was requested (e.g., the Result Page).

Step 9: Response

The web server creates an HTTP response containing the HTML page, CSS files, images, data, JavaScript, and a status code (200 OK).

Step 10: Browser Displays the Page

The response travels back through the web server, router, Internet, ISP, modem, and computer. The browser reads the HTML, applies CSS, executes JavaScript, displays images, and shows the requested page.

3. Internet Protocol (IP) Addresses

An **Internet Protocol (IP) Address** is a unique numerical identifier assigned to every device connected to a network that uses the Internet Protocol. It enables devices to identify each other and communicate over the Internet or a local network — just as a postal address helps deliver a letter to the correct house, an IP address helps deliver data to the correct device.

Definition: An IP Address is a unique logical address assigned to a device on a network to identify it and enable communication using the Internet Protocol.

Versions of IP Addresses

1. IPv4 (Internet Protocol Version 4)

The most widely used version. Uses a 32-bit address divided into four groups called octets. Format example: 192.168.1.10

- 32-bit address
- Four decimal numbers separated by dots
- Each number ranges from 0 to 255
- Supports approximately 4.3 billion unique addresses

2. IPv6 (Internet Protocol Version 6)

Introduced to overcome the shortage of IPv4 addresses. Uses a 128-bit address, giving a much larger address space. Format example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

- 128-bit address

- Eight groups of hexadecimal numbers separated by colons
- Supports an extremely large number of unique addresses
- Provides improved security, efficiency, and scalability

Types of IP Addresses

1. Public IP Address

Assigned by an ISP and accessible over the Internet, allowing devices to communicate with systems outside the local network. Used for hosting websites, Internet communication, online gaming, and email servers.

2. Private IP Address

Used within a local network (LAN) and not directly accessible from the Internet (e.g., 192.168.1.5). Used in home, office, school, and college networks.

3. Static IP Address

Remains the same over time and is manually assigned by a network administrator or ISP. Suitable for web/mail servers, easy to access remotely, and reliable for hosting services.

4. Dynamic IP Address

Automatically assigned by a DHCP (Dynamic Host Configuration Protocol) server and may change when the device reconnects. Easy to manage, efficient use of addresses, and commonly used for home Internet connections.

4. Domain Names

A **Domain Name** is a human-readable name assigned to a website on the Internet that represents the numerical IP address of a web server, so users don't need to remember complex IP addresses (e.g., www.google.com instead of 142.250.193.78).

Definition: A Domain Name is a unique, easy-to-read name used to identify a website on the Internet. It is translated into an IP address by the Domain Name System (DNS).

Need for Domain Names

- IP addresses are difficult for humans to remember
- Domain names provide a simple and meaningful way to access websites
- They uniquely identify websites on the Internet
- They make web browsing easier and more user-friendly

Domain Name System (DNS)

- A hierarchical, distributed database system that translates domain names into IP addresses
- Functions like a phonebook of the Internet
- Process: when you enter www.example.com, DNS finds its IP address so your browser can connect to the server

Structure of a Domain Name

A domain name is structured hierarchically, with parts separated by dots, and is read from right to left — from the most general part to the most specific. For example, in www.google.com:

- **Top-Level Domain (TLD):** the last part of the name (e.g., .com, .org, .edu, or country codes like .uk, .jp) — indicates the purpose or geographic location of a site.
- **Second-Level Domain (SLD):** the part directly left of the TLD — typically the unique brand/site name, such as “google” in google.com.
- **Subdomain:** placed to the left of the SLD to organize sections of a website, e.g., “www”, or mail.google.com and docs.google.com.

How Domain Names Work

- The user enters a domain name such as www.google.com
- The browser sends a request to the DNS
- The DNS server converts the domain name into its corresponding IP address (e.g., www.google.com → 142.250.193.78)
- The browser uses the IP address to locate the web server
- The web server sends the requested webpage to the browser
- The browser displays the webpage to the user

5. World Wide Web (WWW)

The **World Wide Web (WWW)** is a system of interconnected web pages, websites, and online resources accessed over the Internet using a web browser. It allows users to view and share text, images, audio, video, and other multimedia content, using HTTP/HTTPS protocols for communication between browsers and servers. The WWW was invented by Tim Berners-Lee in 1989.

Definition: The WWW is a collection of interconnected web pages and websites accessed through the Internet using web browsers and web servers.

Components of the World Wide Web

1. Web Browser

A software application used to access and display web pages. It sends requests to web servers and displays the received content (e.g., Chrome, Firefox, Edge, Safari).

2. Web Server

Stores website files and responds to browser requests by sending the requested web pages, processing client requests, and hosting websites.

3. Web Page

A single document available on the WWW, usually created using HTML and may include text, images, videos, and hyperlinks (e.g., home page, contact page, about page).

4. Website

A collection of related web pages stored under a single domain name (e.g., college website, shopping website, banking website).

5. URL (Uniform Resource Locator)

The address used to locate a web page, containing the protocol, domain name, and resource/page — e.g., https://www.example.com/index.html

6. HTTP/HTTPS

Communication protocols used to transfer web pages between browsers and servers. HTTP transfers data without encryption; HTTPS transfers encrypted and secure data.

6. Web Browser

A **Web Browser** is a software application that enables users to access, retrieve, interpret, and display web pages and other web resources on the Internet, acting as an interface between the user and the web server by sending HTTP/HTTPS requests and displaying the received pages (e.g., Chrome, Firefox, Edge, Safari, Opera).

Components of a Web Browser

1. User Interface (UI)

The visible part of the browser used to interact with it, including the address bar, back/forward/refresh/home buttons, tabs, bookmarks, and menu.

2. Browser Engine

Acts as a bridge between the user interface and the rendering engine — manages browser operations and coordinates the loading and display of web pages.

3. Rendering Engine

Reads HTML, CSS, and other content and converts it into the visual webpage — parses HTML, applies CSS styles, and updates the page dynamically.

4. Networking Module

Handles communication between the browser and web servers — sends HTTP/HTTPS requests, receives responses, and downloads webpages and files.

5. JavaScript Engine

Executes JavaScript code embedded in web pages, supporting dynamic pages and handling user interactions such as clicks and form validation.

6. Data Storage

Stores user-related information locally, including cookies, cache, browsing history, bookmarks, local storage, and session storage.

7. Web Servers

A **Web Server** is a computer system or software application that stores, processes, and delivers web pages to users over the Internet or a local network. It receives requests from browsers using HTTP/HTTPS and sends back the requested resources, acting as an intermediary between the client and the website.

Definition: A Web Server is a hardware system and software application that receives requests from clients, processes them, and sends the requested web resources using HTTP or HTTPS protocols.

Web Server Operation

Step 1: User Enters the URL

The user enters a website address (e.g., `www.college.edu`) or clicks a hyperlink. The browser prepares an HTTP/HTTPS request for the page.

Step 2: DNS Resolves the Domain Name

The browser sends the domain name to the DNS server, which converts it into its corresponding IP address (e.g., `www.college.edu` → `192.168.10.25`).

Step 3: Browser Sends HTTP/HTTPS Request

Using the IP address, the browser connects to the web server and sends a request, e.g., `GET /index.html HTTP/1.1, Host: www.college.edu`

Step 4: Web Server Receives the Request

The server examines the requested URL, HTTP method, request headers, and cookies/session information.

Step 5: Web Server Processes the Request

For a static webpage, the server retrieves the HTML file directly. For a dynamic webpage, it executes server-side programs (PHP, Python, Java, Node.js) and may query a database.

Step 6: Database Processing (If Required)

For dynamic sites, the server communicates with the database server to retrieve or update information such as exam results, customer data, product details, or banking records.

Step 7: Web Server Creates HTTP Response

The server prepares a response with a status code (200 OK, 404 Not Found, etc.), response headers, HTML page, images, CSS, JavaScript, and other resources.

Step 8: Response Sent to Browser

The HTTP response travels through the Internet back to the client's browser.

Step 9: Browser Displays the Webpage

The browser reads the HTML, applies CSS, executes JavaScript, loads media, and displays the complete webpage.

General Characteristics of a Server

- **High Performance:** powerful processors, large memory, and high-speed storage for quick responses.
- **High Availability:** operates 24×7 with minimal downtime.
- **Multi-User Support:** serves many clients simultaneously.
- **Reliability:** built for continuous, uninterrupted operation.
- **Scalability:** can be upgraded with more processors, memory, storage, or additional servers.
- **Security:** authentication, password protection, access control, firewalls, encryption, and HTTPS.
- **Data Storage:** large capacity for files, databases, applications, websites, emails, and backups.
- **Resource Sharing:** shares files, printers, databases, applications, and Internet connections.
- **Backup and Recovery:** regular backups allow data restoration after failure or attack.
- **Fault Tolerance:** continues functioning even if a component fails (RAID, redundant power, etc.).
- **Remote Access:** can be monitored and managed remotely.
- **Network Connectivity:** connects to high-speed networks supporting TCP/IP, HTTP, HTTPS, FTP, and SMTP.

8. Uniform Resource Locator (URL)

A **URL** is the address of a resource on the Internet. It tells the web browser where a resource is located and how to access it using a protocol such as HTTP or HTTPS. Example:
`https://www.example.com/products/laptop.html`

Need for a URL

- Identifies the location of a web resource
- Specifies the protocol used for communication
- Helps browsers retrieve web pages, images, videos, and documents
- Enables users to access resources from anywhere on the Internet
- Provides a unique address for every web resource

General Structure of a URL

Protocol://Domain Name:Port/Path/Filename?Query String#Fragment

Example:
`https://www.amazon.in:443/electronics/mobile.html?brand=samsung#reviews`

Components of a URL

Component	Example	Description
Protocol	https	Specifies how communication takes place between browser and server.
Domain Name	www.example.com	Human-readable name of the website.
Port Number	443	Indicates the communication port (usually optional).
Path	/folder/	Location of the resource on the server.
File Name	page.html	Name of the webpage or file.
Query String	?id=100	Sends additional information to the server.
Fragment	#contact	Jumps to a particular section within the webpage.

How a URL Works

- The user enters a URL in the browser
- The browser identifies the protocol (HTTP or HTTPS)
- The browser asks the DNS server for the IP address of the domain
- DNS returns the IP address
- The browser establishes a connection with the web server
- The browser sends an HTTP request
- The web server processes the request
- The server sends an HTTP response containing the requested webpage
- The browser renders and displays the webpage

9. MIME (Multipurpose Internet Mail Extensions)

MIME is a standard that extends the format of Internet messages, allowing the transfer of different data types such as text, images, audio, video, PDF files, and application files. In simple terms, MIME tells the

browser what kind of file it has received and how to handle or display it.

Why is MIME Needed?

Initially, email systems could send only plain text (ASCII). They could not send images, videos, documents, or other multimedia files. MIME was introduced to support multimedia content, allow file attachments, specify content types, and enable browsers to correctly interpret downloaded files.

Features of MIME

- Supports text, images, audio, video, and application files
- Allows email attachments
- Used in HTTP communication between browsers and web servers
- Specifies the content type of files
- Platform-independent
- Supports multiple character sets (UTF-8, ASCII, etc.)

Advantages of MIME

- Supports multimedia content and file attachments in emails
- Helps browsers identify file types and render web content correctly
- Supports various character encodings and is platform-independent

Limitations of MIME

- Incorrect MIME types may cause files to be displayed incorrectly
- Some older email systems do not fully support all MIME features
- MIME alone does not provide security or encryption

10. HTTP Request and Response

HTTP (HyperText Transfer Protocol) is the protocol used for exchanging information between a web browser (client) and a web server. When a user visits a website, the browser sends an HTTP Request, and the server sends back an HTTP Response containing the requested resource (HTML page, image, video, or JSON data). HTTP follows a client-server model.

- **HTTP Request:** a message sent by the client to the web server asking for a specific resource or service.
- **HTTP Response:** a message sent by the server back to the client, containing the requested resource or information about the request.

HTTP Request Phase

The phase in which the browser sends a request to the web server for a resource such as a web page, image, video, or file.

Step 1: User Enters the URL

The browser identifies the protocol, domain name, and requested resource (e.g., `https://www.college.edu/index.html`).

Step 2: DNS Lookup

The browser contacts DNS to convert the domain name into its IP address (e.g., `www.college.edu` → `192.168.10.20`).

Step 3: Establish Connection

The browser connects to the web server using the IP address. HTTP uses TCP port 80; HTTPS uses TCP port 443.

Step 4: Browser Creates the HTTP Request

The request includes the request method (GET, POST, etc.), requested resource, HTTP version, headers, and body if required.

Step 5: Browser Sends the Request

The request travels through the Internet to the web server.

Step 6: Web Server Receives the Request

The server checks which resource is requested, whether it exists, and whether the user has permission to access it.

Components of an HTTP Request

Component	Description
Request Line	Contains method, URL, and HTTP version
Headers	Provide additional information (Host, User-Agent, Accept, etc.)
Blank Line	Separates headers from the body
Request Body	Contains data (used with POST, PUT, etc.)

Common HTTP Request Methods

Method	Purpose
GET	Retrieve data from the server
POST	Send data to the server
PUT	Update existing data
DELETE	Delete a resource
HEAD	Retrieve only response headers
OPTIONS	List supported request methods
PATCH	Partially update a resource

HTTP Response Phase

The phase in which the web server processes the client's request and sends an HTTP Response back to the browser, containing the requested resource or an appropriate status message.

Step 1: Server Receives the Request

The server identifies the requested resource, request method, and client information from the headers.

Step 2: Server Processes the Request

The server checks whether the file exists, whether the user has permission, and whether server-side processing (PHP, JSP, ASP.NET, etc.) is required.

Step 3: Server Creates the HTTP Response

The response contains a status line, response headers, a blank line, and a response body.

Step 4: Server Sends the Response

The response travels through the Internet to the client's browser.

Step 5: Browser Receives and Displays the Resource

The browser interprets the data and displays it — e.g., HTML as a webpage, JPEG as an image, PDF in a viewer, MP4 in a video player.

Components of an HTTP Response

- **Status Line:** contains the HTTP version, status code, and status message (e.g., HTTP/1.1 200 OK).
- **Response Headers:** provide additional information such as content type, content length, server details, and cookies (e.g., Content-Type: text/html).
- **Blank Line:** separates the response headers from the response body.
- **Response Body:** contains the actual data requested, such as an HTML page, image, PDF, JSON, or video.

11. Security Issues in World Wide Web (WWW)

Security issues in the WWW are threats or attacks that affect the security of websites, web servers, and users, potentially leading to data theft, unauthorized access, loss of information, financial loss, or disruption of services. As more services such as banking, shopping, education, and social networking move online, web security has become very important.

1. Malware

Software intentionally created to harm a computer, steal information, or disrupt normal functioning. It can enter through infected websites, email attachments, or downloaded files. Common types include viruses, worms, Trojan horses, ransomware, and spyware.

2. Phishing

A cyberattack in which attackers create fake websites, emails, or messages appearing to be from trusted organizations, aiming to trick users into revealing sensitive information such as usernames, passwords, bank details, or OTPs.

3. Hacking

The act of gaining unauthorized access to a computer system, network, or website in order to access, modify, steal, or destroy information without the owner's permission.

4. Password Attack

An attempt to obtain or guess a user's password to access an account without authorization, often using software tools or common password lists.

5. SQL Injection

A web security attack in which malicious SQL commands are entered into a website's input fields (such as login forms). If input isn't properly validated, the attacker can access, modify, or delete data in the database.

6. Cross-Site Scripting (XSS)

An attack in which malicious JavaScript code is inserted into a trusted website. When users visit the affected page, the script runs in their browser and may steal cookies, session information, or personal data.

7. Denial of Service (DoS/DDoS)

A DoS attack occurs when an attacker sends a large number of requests to a server, making it too busy to respond to genuine users. A DDoS attack is similar but uses many infected computers simultaneously.

8. Data Theft

The unauthorized stealing or copying of confidential information from a website, database, or computer system, such as personal details, passwords, or financial records.

Security Measures

- Use HTTPS instead of HTTP
- Install antivirus software
- Use strong and unique passwords
- Enable Two-Factor Authentication (2FA)
- Keep browsers and software updated
- Install firewalls
- Avoid clicking unknown links
- Validate user input in web applications
- Take regular backups of important data

12. Introduction to HTML

HTML (HyperText Markup Language) is the standard markup language used to create and design web pages. It provides the basic structure of a webpage using tags to organize and display content such as text, images, tables, links, forms, audio, and videos.

HTML is not a programming language because it does not perform calculations or make decisions — it is a markup language that tells the browser how to display content.

Definition: HTML is the standard markup language used to create and structure web pages on the World Wide Web, using predefined tags to define layout and presentation.

Breaking Down the Name

- **Hyper:** refers to hyperlinks, which connect one web page to another, letting users navigate by clicking text or images.
- **Text:** HTML is mainly used to display text — headings, paragraphs, lists, tables, and other content.
- **Markup:** adding tags to content to describe its structure and appearance, e.g., the `<b>` tag displays text in bold.
- **Language:** HTML follows a set of rules (syntax) and uses predefined tags to build web pages.

Features of HTML

- Simple and easy to learn
- Platform-independent (Windows, Linux, macOS, etc.)
- Supported by all modern web browsers

- Uses tags to structure web pages
- Supports multimedia such as images, audio, and videos
- Allows hyperlinks to connect different web pages
- Can be combined with CSS for styling and JavaScript for interactivity

Applications of HTML

HTML is used to create personal, educational, business, and e-commerce websites, blogs and news portals, online forms, and web applications.

Advantages and Limitations

- **Advantages:** easy to write and understand, free and open standard, supported by all browsers, integrates easily with CSS and JavaScript, and helps create structured web pages.
- **Limitations:** cannot perform calculations or logical operations, cannot create dynamic pages by itself, requires CSS for attractive design, and requires JavaScript for interactive features.

Standard Structure of an HTML Document

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Web Page</title>
  </head>
  <body>
    <h1>Welcome to HTML</h1>
    <p>This is my first web page.</p>
    
    <a href="https://www.example.com">Visit Example</a>
  </body>
</html>
```

- **<!DOCTYPE html>:** tells the browser the document is an HTML5 document; it is the first line of every HTML document and is not itself an HTML tag.
- **<html> ... </html>:** the root tag of an HTML document; all other tags are written inside it, marking the beginning and end of the webpage.
- **<head> ... </head>:** contains information about the webpage (title, meta tags, CSS, and JavaScript links) that is not visible on the page.
- **<title> ... </title>:** specifies the title of the webpage, shown on the browser tab; written inside <head>.
- **<body> ... </body>:** contains all the visible content of the webpage — headings, paragraphs, images, links, tables, lists, forms, etc.

Components of HTML

1. Elements

An HTML element is the complete part of a document, including the opening tag, content, and closing tag. Syntax: <tagname>Content</tagname>. For example, the entire <p>Welcome to HTML</p> is an HTML element.

2. Tags

Keywords enclosed within angle brackets (< >) that tell the browser how to display content. Most tags are written in pairs (opening and closing), though some, such as
 and <hr>, are self-closing (empty) tags that need no closing tag.

3. Attributes

Attributes provide additional information about an element. They are written inside the opening tag as name-value pairs, e.g., in `<a href="https://www.google.com">Google</a>`, href is an attribute specifying the link's destination.

Types of HTML Elements

- **Block-level Elements:** start on a new line and occupy the full available width, e.g., `<h1>`–`<h6>`, `<p>`, `<div>`, `<table>`, `<form>`, `<ul>`, `<ol>`.
- **Inline Elements:** do not start on a new line and occupy only the space required, mostly used for formatting text, e.g., `<b>`, `<i>`, `<u>`, `<span>`, `<a>`, `<img>`, `<strong>`, `<em>`.
- **Self-Closing (Empty) Elements:** have only an opening tag and no content, e.g., `<br>`, `<hr>`, `<img>`, `<input>`, `<meta>`, `<link>`.
- **Nested Elements:** elements placed inside another element (a child inside a parent), used to organize webpage content, e.g., a `<h2>` and `<p>` nested inside a `<div>`.

HTML Attributes

Attributes are special properties, written inside the opening tag as `name="value"`, that provide additional information about an element or modify its behavior or appearance. An element can have multiple attributes, and values should be enclosed in double quotes.

1. Core (Global) Attributes

Attribute	Purpose
id	Gives a unique identifier to an element.
class	Groups elements for CSS and JavaScript.
title	Displays additional information as a tooltip.
style	Applies inline CSS styles.
lang	Specifies the language of the content.

2. Element-Specific Attributes

Element	Attribute	Purpose
<code><a></code>	href	Specifies the destination URL.
<code></code>	src	Specifies the image location.
<code></code>	alt	Displays alternative text if the image cannot be loaded.
<code><input></code>	type	Specifies the type of input field.
<code><table></code>	border	Specifies the table border (obsolete in HTML5).

3. Boolean Attributes

Boolean attributes have only two states: present or absent. If present, the feature is enabled, e.g., checked, disabled, readonly, required, selected.

```
<input type="checkbox" checked>
<input type="text" readonly>
```

```
<input type="submit" disabled>
```

4. Event Attributes

Event attributes execute JavaScript when a specific event occurs, such as clicking a button or moving the mouse, e.g., onclick, onmouseover, onchange, onkeyup.

```
<button onclick="alert('Welcome!')">Click Me</button>
```

13. HTML Headings

HTML headings define the titles and subheadings of a web page, organizing content into sections for readability and improving SEO. HTML provides six levels, <h1> (largest, most important, usually the main title, used once per page) through <h6> (smallest, least important).

```
<!DOCTYPE html>
<html>
<head><title>HTML Headings</title></head>
<body>
<h1>Government First Grade College</h1>
<h2>Department of Computer Applications</h2>
<h3>BCA Programme</h3>
</body>
</html>
```

14. HTML Paragraph

An HTML paragraph displays blocks of text and is created using the <p> tag. The browser automatically adds spacing before and after each paragraph, and ignores extra spaces and line breaks inside it.

```
<p>Welcome to HTML.</p>
<p>HTML is used to create web pages.</p>
```

15. HTML Text Formatting

HTML text formatting tags change the appearance or emphasis of text, such as making it bold, italic, underlined, highlighted, or displaying superscripts and subscripts.

Tag	Purpose
	Displays text in bold.
	Indicates important text (usually bold).
<i>	Displays text in italic.
	Emphasizes text (usually italic).
<u>	Underlines text.
<mark>	Highlights text.
<small>	Displays smaller text.
	Shows deleted (strikethrough) text.

<code><ins></code>	Shows inserted (underlined) text.
<code><sup></code>	Displays superscript text.
<code><sub></code>	Displays subscript text.

`Bold <i>Italic</i> <u>Underline</u> <mark>Highlight</mark>
H₂O x²`

16. HTML Hyperlinks

A hyperlink is clickable text or an image that connects one webpage to another webpage, file, email address, or website, created using the `<a>` (anchor) tag.

```
<a href="URL">Link Text</a>
<a href="https://www.google.com" target="_blank">Google</a>
```

Types of Hyperlinks

- External Link – links to another website
- Internal Link – links to another page within the same website
- Email Link – opens the email application
- Bookmark Link – links to a specific section of the same webpage

17. HTML Images

The `<img>` tag displays images on a webpage. It is an empty (self-closing) element with no closing tag.

```

```

Advantages of Using alt

- Displays text if the image cannot be loaded
- Improves accessibility for visually impaired users
- Helps search engines understand the image

18. Preserving White Spaces in HTML

By default, HTML ignores extra spaces and line breaks. To display text exactly as written — including all spaces and line breaks — use the `<pre>` tag. This is commonly used for code, poems, and formatted text.

```
<pre>
Name   : Ravi
Course : BCA
College: GFGC
</pre>
```

19. HTML Lists

HTML lists display a group of related items in an organized manner. HTML provides three types of lists — Ordered, Unordered, and Description — and also allows nested lists, where one list is placed inside

another.

1. Ordered List ()

Displays items in a specific sequence, automatically numbered or lettered. Used when the order of items matters.

```
<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Attributes of

Attribute	Purpose
type	Sets numbering style: 1, A, a, I, or i (e.g., type="A" → A, B, C).
start	Starts numbering from a specified value (e.g., start="5").
reversed	Displays the list in reverse order.

2. Unordered List ()

Displays items using bullets instead of numbers, used when the order of items is not important.

```
<ul>
  <li>Keyboard</li>
  <li>Mouse</li>
  <li>Monitor</li>
</ul>
```

The type attribute (or CSS in HTML5) sets the bullet style: disc, circle, or square.

3. Description List (<dl>)

Displays terms and their descriptions, commonly used for glossaries or question-answer content, using <dl> (list), <dt> (term), and <dd> (description).

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Nested Lists

A nested list is placed inside another list, used to represent categories and subcategories — for example, an unordered list of programming languages nested inside an ordered list of topics.

```
<ol>
  <li>Programming Languages
    <ul>
      <li>C</li>
      <li>Java</li>
      <li>Python</li>
    </ul>
  </li>
</ol>
```

```

    </li>
    <li>Web Technologies
      <ul>
        <li>HTML</li>
        <li>CSS</li>
      </ul>
    </li>
  </ol>

```

20. HTML Tables

An HTML table organizes and displays data in rows and columns, useful for structured information such as student details, marks, employee records, timetables, and product lists. It is created using the `<table>` tag.

Structure of an HTML Table

Tag	Purpose
<code><table></code>	Creates the table.
<code><tr></code>	Defines a table row.
<code><th></code>	Defines a table header cell.
<code><td></code>	Defines a table data cell.
<code><caption></code>	Adds a title to the table.

Example

```

<table border="1">
  <caption>Student Details</caption>
  <tr>
    <th>USN</th><th>Name</th><th>Course</th>
  </tr>
  <tr>
    <td>101</td><td>Rahul</td><td>BCA</td>
  </tr>
  <tr>
    <td>102</td><td>Anita</td><td>BCA</td>
  </tr>
</table>

```

HTML Table Attributes

Attribute	Purpose	Example
border	Displays a border around the table.	border="1"
cellpadding	Adds space inside each cell.	cellpadding="10"
cellspacing	Adds space between cells.	cellspacing="5"
width	Sets the width of the table.	width="80%"

align	Aligns the table (left/center/right).	align="center"
bgcolor	Sets the background color.	bgcolor="lightblue"

Cell Spanning

- **colspan:** merges two or more columns into one cell.
- **rowspan:** merges two or more rows into one cell.

```
<th colspan="2">Student Details</th>
<th rowspan="2">Name</th>
```

21. The <div> Element

The <div> (Division) element is a block-level container used to divide a webpage into sections, group related content, apply CSS styles, and add JavaScript functionality. It has no visual effect unless styles are applied.

```
<div style="background-color:lightblue; padding:10px;">
  <h2>Student Information</h2>
  <p>Name: Rahul</p>
</div>
```

Difference Between <div> and

<div>	
Block-level element	Inline element
Starts on a new line	Does not start on a new line
Occupies full width	Occupies only required width
Used to group large sections	Used to group small portions of text

22. Horizontal Rule (<hr>)

The <hr> tag draws a horizontal line on a webpage to separate different sections or topics, improving organization and readability.

```
<h2>Introduction</h2>
<p>HTML is a markup language.</p>
<hr>
<h2>Conclusion</h2>
<p>HTML is used to create web pages.</p>
```

23. HTML Character Entities

Some characters, such as <, >, and &, have special meaning in HTML. Character entities are used to display these reserved characters, as well as symbols that cannot be typed directly, such as the copyright symbol (©), trademark (™), and non-breaking spaces.

Syntax: &entity_name; or &#entity_number;

Common HTML Character Entities

Character	Entity Name	Entity Number
<	<	<
>	>	>
&	&	&
"	"	"
'	'	'
Space	 	
©	©	©
®	®	®
™	™	™
■	₹	₹